

11.03.26
Wednesday

Module - 5 Correlation and Regression

If 2 quantity vary in such a way that movements in 1 are accompanied by movements in the other, then the quantities are correlated.

eg1: An increase in the amount of rain will be accompanied by an increase in sales of umbrellas.

The correlation is then called positive correlation.

eg2: An increase in the price of a commodity will ~~decrease~~ result in decrease in sales.

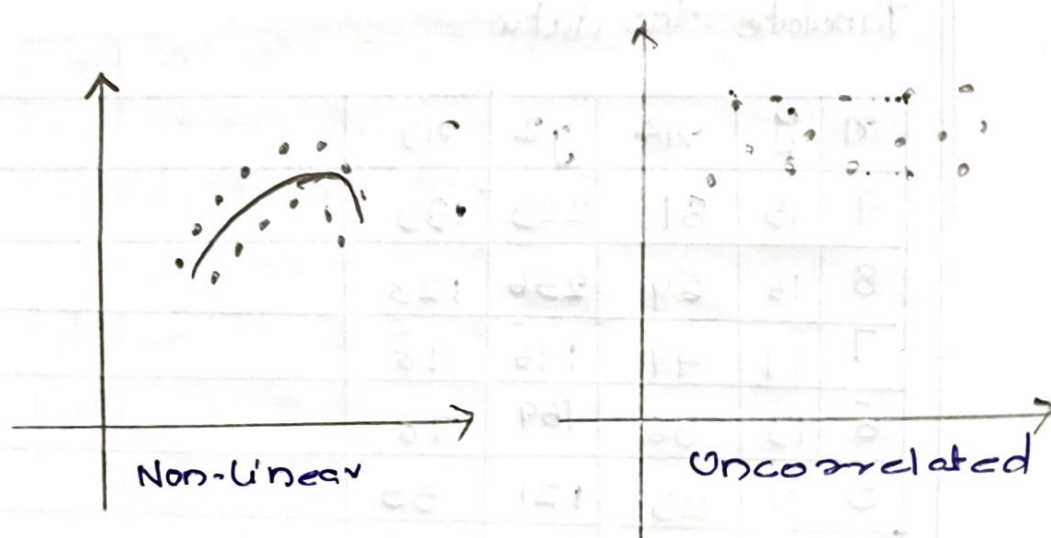
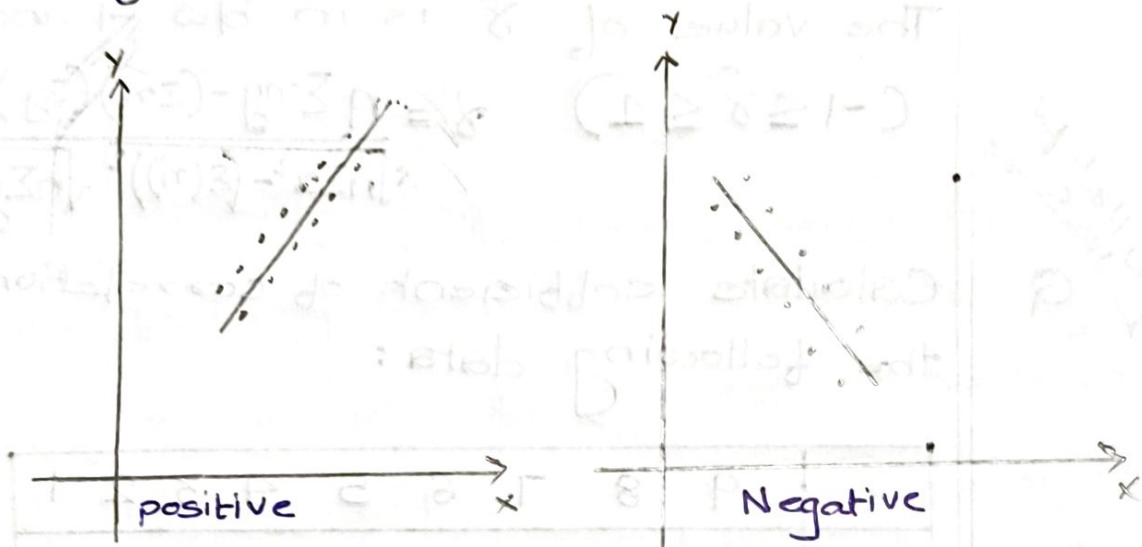
This correlation is then called Negative correlation

Also, there may be no-connection b/w commodities even if they just happen to be correlated.

Eg: Increase in rain in India and wheat production in USA.

Scatter diagram.

A useful method of investigating if there is any correlation b/w 2 variables is to draw a scatter diagram in respect of each observation ordered pair (x, y) plot the point in any plane. Such a diagram is called scatter-diagram



The Coefficient of Correlation.

It is the measurement of the degree of correlation b/w variables. Most widely used is Karl Pearson's ^{coefficient of} correlation.

and it is denoted by γ or γ_{xy} .

Equation:

$$\gamma = \frac{\overline{xy} - \bar{x}\bar{y}}{\sqrt{\frac{1}{n} \sum (x - \bar{x})^2} \cdot \sqrt{\frac{1}{n} \sum (y - \bar{y})^2}} = \frac{\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})}{\sqrt{\frac{1}{n} \sum (x - \bar{x})^2} \cdot \sqrt{\frac{1}{n} \sum (y - \bar{y})^2}}$$

The value of γ is in b/w -1 and 1

$$(-1 \leq \gamma \leq 1) \quad \gamma = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

✓ will be for
series

Calculate coefficient of correlation from the following data:

x	9	8	7	6	5	4	3	2	1
y	15	16	14	13	11	12	10	8	9

Tabulate the data

x	y	x ²	y ²	xy	
9	15	81	225	135	
8	16	64	256	128	
7	14	49	196	98	
6	13	36	169	78	
5	11	25	121	55	
4	12	16	144	48	
3	10	9	100	30	
2	8	4	64	16	
1	9	1	81	9	
Σ	45	108	285	1356	594

n = no. of observation of any one variable
 $= 9$

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{n \sum x^2 - (\sum x)^2} \cdot \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$= \frac{9 \times 597 - 45 \times 108}{\sqrt{9 \times 285 - 2025} \cdot \sqrt{9 \times 1356 - 11664}}$$

$$= \cancel{0.48} \underline{\underline{0.950}}$$

✓ Q2. Compute the correlation coefficient from the following data.

x : 77 54 27 52 14 35 90 25 56 60
 y : 35 58 60 40 50 40 35 56 34 42

$r = -0.665$

Tabulate the data:

x	y	x^2	y^2	xy
77	35	5929	1225	2695
54	58	2916	3364	3132
27	60	729	3600	1620
52	40	2704	1600	2080
14	50	196	2500	700
35	40	1225	1600	1400
90	35	8100	1225	3150
25	56	625	3136	1400
56	34	3136	1156	1904
60	42	3600	1764	2520
Σ	460	450	29160	20170

$n = 10$

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{n \sum x^2 - (\sum x)^2} \cdot \sqrt{n \sum y^2 - (\sum y)^2}}$$

$$r = \frac{10 \times 19851 - (460 \cdot 450)}{\sqrt{10 \cdot 29160 - (460)^2} \cdot \sqrt{10 \cdot 20170 - 202500}}$$

$$r = 198510 - (00)$$

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Regression analysis

studies the relationship b/w variables.

- x: Independent variable
- y: Dependent variable
- Used to predict values of y from x

Linear regression equation:

$$y = a + bx$$

Regression of y on x

$$y = a + bx$$

Regression of x on y:

$$x = c + dy$$

The two regression lines are generally different.

Least Square Method : to obtain Regression Line.

Principle : Minimize the sum of squared deviation.

For $y = a + bx$

Normal equations are:

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

Regression line through mean:

$$\bar{y} = a + b\bar{x}$$

$$y - \bar{y} = b(x - \bar{x})$$

Regression line passes through $(\bar{x}, \bar{y})$

'b' is called regression coefficient.

$$b = \frac{\sigma_{xy}}{\sigma_x^2}$$

Regression equation:

$$y - \bar{y} = \frac{\sigma_{xy}}{\sigma_x^2} (x - \bar{x}) \quad \text{--- (1)}$$

Correlation coefficient :

$$r = \frac{\sigma_{xy}}{\sigma_x \cdot \sigma_y}$$

$$\sigma_{xy} = r \cdot \sigma_x \cdot \sigma_y \quad \text{--- (2)}$$

~~Sub (2) in (1)~~

$$y - \bar{y} = r$$

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

also Regress line through mean of x and y

$$x - \bar{x} = \frac{r \sigma_x}{\sigma_y} (y - \bar{y})$$

$$b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

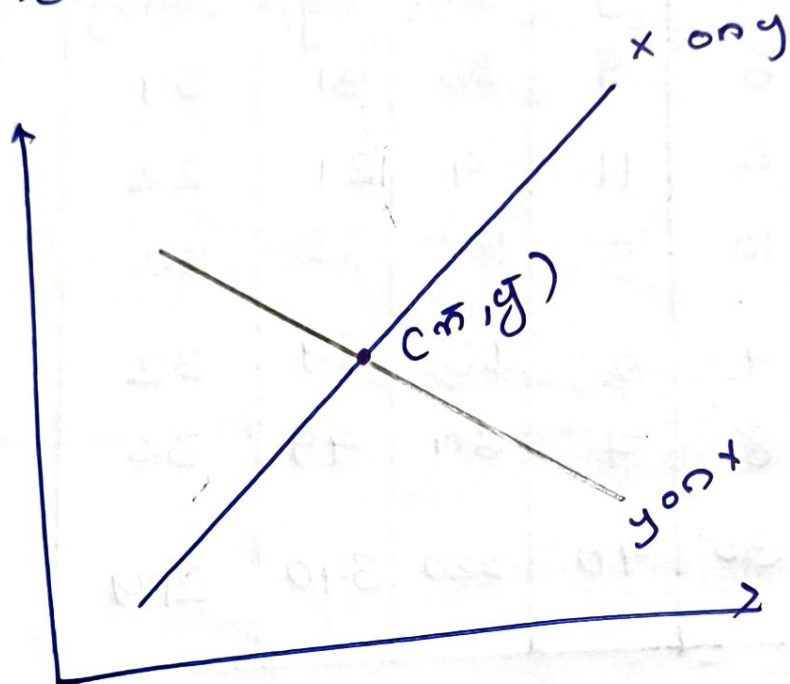
$$\boxed{b_{yx} \cdot b_{xy} = r^2}$$

$$\therefore r = \pm \sqrt{b_{yx} \cdot b_{xy}}$$

Properties

- Both line passes through $(\bar{x}, \bar{y})$
- If $r = \pm 1$, both lines coincide.
- If $r = 0$, regression lines become $y = \bar{y}$.
- product of regression coefficient is r^2 .

Geometric Interpretation



From the following data obtain 2 regression Equations:

x	y
6	9
2	11
10	5
4	8
8	7

The normal equations are:

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

n = no. of observation

x	y	x^2	y^2	xy
6	9	36	81	54
2	11	4	121	22
10	5	100	25	50
4	8	16	64	32
8	7	64	49	56
Σ	30	40	220	340
				214

$\Rightarrow$ Sub:

$$40 = 5a + b \times 30$$

$$\Rightarrow 40 = 5a + 30b$$

$$214 = a \times 30 + b \times 220$$

$$\Rightarrow 40 = 5a + 30b \quad \times 6$$

$$214 = 30a + 220b$$

$$240 = 30a + 40b$$

$$(c) \ 214 = 30a + 220b$$

$$26 = -40b$$

$$b = \frac{-13}{20}$$

$$\boxed{b = -0.65}$$

$$40 = 5a + (-19.5)$$

$$59.5 = 5a$$

$$\boxed{11.9 = a}$$

$\therefore$ The regression line y on x is

$$\boxed{y = 11.9 + (-0.65)x}$$

Now regression line x on y :

$$\sum x = nc + d \sum y$$

$$\sum xy = c \sum y + d \sum y^2$$

$$\Rightarrow 30 = 5c + 40d$$

$$214 = 40c + 340d$$

$$\Rightarrow 30 = 5c + 40d \quad \times 8$$

$$214 = 40c + 340d$$

$$\Rightarrow \cancel{240} = 40c + 320d$$

$$240 = 40c + 320d$$
$$e) 214 = 40c + 340d$$

$$26 = -20d$$

$$\frac{-13}{10} = d$$

$$\boxed{d = -1.3}$$

$$30 = 5c + 40 \times (-1.3)$$

$$30 = 5c - 52$$

$$82 = 5c$$

$$\boxed{c = 16.4}$$

$\therefore$ The regression line of x on y

$$\boxed{x = 16.4 - 1.3y}$$

Method : 2

The regression of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$b_{yx} = \frac{\sigma_{xy}}{\sigma_x^2}$$

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{30}{5}$$

$$= \underline{\underline{6}}$$

$$\bar{y} = \frac{40}{5}$$

$$= \underline{\underline{8}}$$

begin

$$\sigma_x^2 = \text{variance of } x$$

$$= \frac{\sum x^2}{n} - \bar{x}^2$$

$$= \frac{220}{5} - 36$$

$$= 44 - 36$$

$$= \underline{\underline{8}}$$

$$\sigma_y^2 = \text{variance of } y$$

$$= \frac{\sum y^2}{n} - \bar{y}^2$$

$$= \frac{340}{5} - 64$$

$$= 68 - 64$$

$$= \underline{\underline{4}}$$

$$\sigma_{xy} = \frac{\sum xy}{n} - (\bar{x} \cdot \bar{y})$$

$$= \frac{814}{5} - 48$$

$$= \underline{\underline{-5.2}}$$

$$y - 8 = (-0.65)(x - 6)$$

$$y = 11.9 - 0.65x$$

Regn of x on y

$$x - \bar{x} = b_{yx}(y - \bar{y})$$

$$x - 6 = \frac{-5.2}{4}(y - 8)$$

$$x - 6 = (-1.3)(y - 8)$$

$$x - 6 = -1.3y + 10.4$$

$$x = 16.4 - 1.3y$$

Ex 100.

Find the coefficient of correlation and the 2 regression line.

x	y
20	31
22	29
25	32
26	37
27	35
23	34

$$n = 6$$

x	y	x ²	y ²	xy
20	31	400	961	620
22	29	484	841	638
25	32	625	1024	800
26	37	676	1369	962
27	35	729	1225	945
23	34	529	1156	782
143	198	3443	6576	4747

Coefficient of correlation r

$$r = \frac{\sigma_{xy}}{\sigma_x \cdot \sigma_y}$$

$$\begin{aligned} \sigma_{xy} &= \frac{\sum xy}{n} - (\bar{x} \cdot \bar{y}) \\ &= \frac{4747}{6} - (23.8 \cdot 33) \\ &= \cancel{791.16} - 5.76 \\ &= \underline{5.7666} \end{aligned}$$

$$\begin{aligned} \sigma_x &= \frac{\sum x}{n} - \bar{x} \\ &= \frac{143}{6} - 23.8 \\ &= \underline{0.33} \\ \sigma_y &= \frac{\sum y}{n} - \bar{y} \\ &= \frac{198}{6} - 33 \\ &= 0 \end{aligned}$$

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Spearman's Rank Correlation

Rank Correlation:

Instead of finding the coefficient of correlation b/w the actual values of the two variables it is often sufficient to find the correlation of the rank of the variable.

Eg: Instead of giving marks one can rank the students according to their merits.

Also rank correlation can be used in circumstances where it is difficult to quantitatively measure the variables.

Eg: Intelligence or beauty.

The Spearman's Rank Correlation is given by:

$$r = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i = difference in ranks of x_i & y_i

The formula is used when there is no tie's in rank.

If there are ties we substitute each of the tied observation, the mean of the rank they jointly occupied. If there are 3 ranks = 4, they are given mean rank = $\frac{4+5+6}{3} = 5$,

If there are 2 ranks = 4, they are given mean rank = $\frac{4+5}{2} = 4.5$.

For each m group of tied rank add a factor $\frac{1}{12} m(m^2-1)$ to $\sum d_i^2$, thus

$$r = 1 - \frac{6 \left[\sum d_i^2 + \frac{1}{12} m_1(m_1^2-1) + \frac{1}{12} m_2(m_2^2-1) + \dots \right]}{n(n^2-1)}$$

Q Calculate the rank correlation for ranks in

x:	1	2	3	4
y:	3	4	2	1

Step 1: each if 2 ranks are same for any ordered pair.

Step 2: form the table

R_m	R_y	$d = R_m - R_y$	d^2
1	3	-2	4
2	4	-2	4
3	2	1	1
4	1	3	9
			$\Sigma d^2 = 18$

$$r = \frac{1 - 6 \Sigma d_i^2}{n(n^2 - 1)}$$

$$= \frac{1 - 6 \times 18}{4(16 - 1)}$$

$$= \frac{1 - 128}{4 \times 15}$$

$$= \frac{1 - 128}{60}$$

$$= \frac{60 - 128}{60}$$

$$= \frac{-68}{60}$$

$$= \frac{-6.8}{6}$$

$$= \underline{\underline{-0.8}}$$

Obtain the rank correlation for the data

x: 68 64 75 50 64 80 75 40 55 64
y: 62 58 68 45 81 60 68 48 50 70

Ans step 1: Rank the data

- Highest value 1st
- descending ord.

x:

80	75	75	68	64	64	64	55	50	40
1	2	2	3	4	4	4	5	6	7
①	②	②	④	⑤	⑤	⑤	⑧	⑨	⑩

y:

81	70	68	68	62	60	58	50	48	45
①	2	3	3	4	5	6	7	8	9
①	②	③	③	⑤	⑥	⑦	⑧	⑨	⑩

In the 'x' series the value 75 occurs twice, had these value have been slightly different they could have been given ranks '2' and '3'. Therefore the common rank is $\text{mean } \frac{2+3}{2} = \underline{\underline{2.5}}$.

The value 64 occurs thrice they would have been given next ranks 5, 6, 7

$\therefore$ common rank $\frac{5+6+7}{3} = \underline{\underline{6}}$

|||¹ in Y-series 68 occurs twice,
they could have been given the mid
3, 4 $\therefore$ mean rank = $\frac{3+4}{2} = \underline{\underline{3.5}}$

Step 3: Table dr.

x	y	R _x	R _y	d	d ²
68	62	4	5	-1	1
64	58	6	7	-1	1
75	64	2.5	3.5	-1	1
50	45	9	10	-1	1
64	81	6	1	5	25
80	60	1	6	-5	25
75	68	2.5	3.5	-1	1
40	48	10	9	1	1
55	50	8	8	0	0
64	70	6	2	4	16
				Σd^2	72

$$\begin{aligned}
 1) \text{Correction factors of 75 } (m_1 = 2) &= \frac{1}{2}(m[m^2 - 1]) \\
 &= \frac{1}{12} \times 2(4 - 1) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 2) \text{Correction factors of 64 } (m_2 = 3) &= \frac{1}{12} \times 3(9 - 1) \\
 &= \frac{1}{12} \times 3 \times 8 \\
 &= \underline{\underline{2}}
 \end{aligned}$$

Correction factor 0.68 ($m_3=2$) = $\frac{1}{2}$

$$\therefore \gamma = \frac{1 - 6 \left[\sum d_i^2 + \frac{1}{12} m_1 (m_1^2 - 1) + \frac{1}{12} m_2 (m_2^2 - 1) + \frac{1}{12} m_3 (m_3^2 - 1) \right]}{n(n^2 - 1)}$$

$$= \frac{1 - 6 \left[72 + \frac{1}{2} + 2 + \frac{1}{2} \right]}{10(10^2 - 1)}$$

$$= \frac{1 - 6 \left[\frac{154}{2} + 1 + 4 + 1 \right]}{10(10^2 - 1)}$$

$$= \frac{1 - 6 \left[\frac{150}{2} \right]}{10(99)}$$

4

$$= 1 - \frac{450}{990}$$

$$= \frac{540}{990}$$

$$= 1 - \frac{450}{990}$$

$$= \underline{\underline{0.545}}$$